

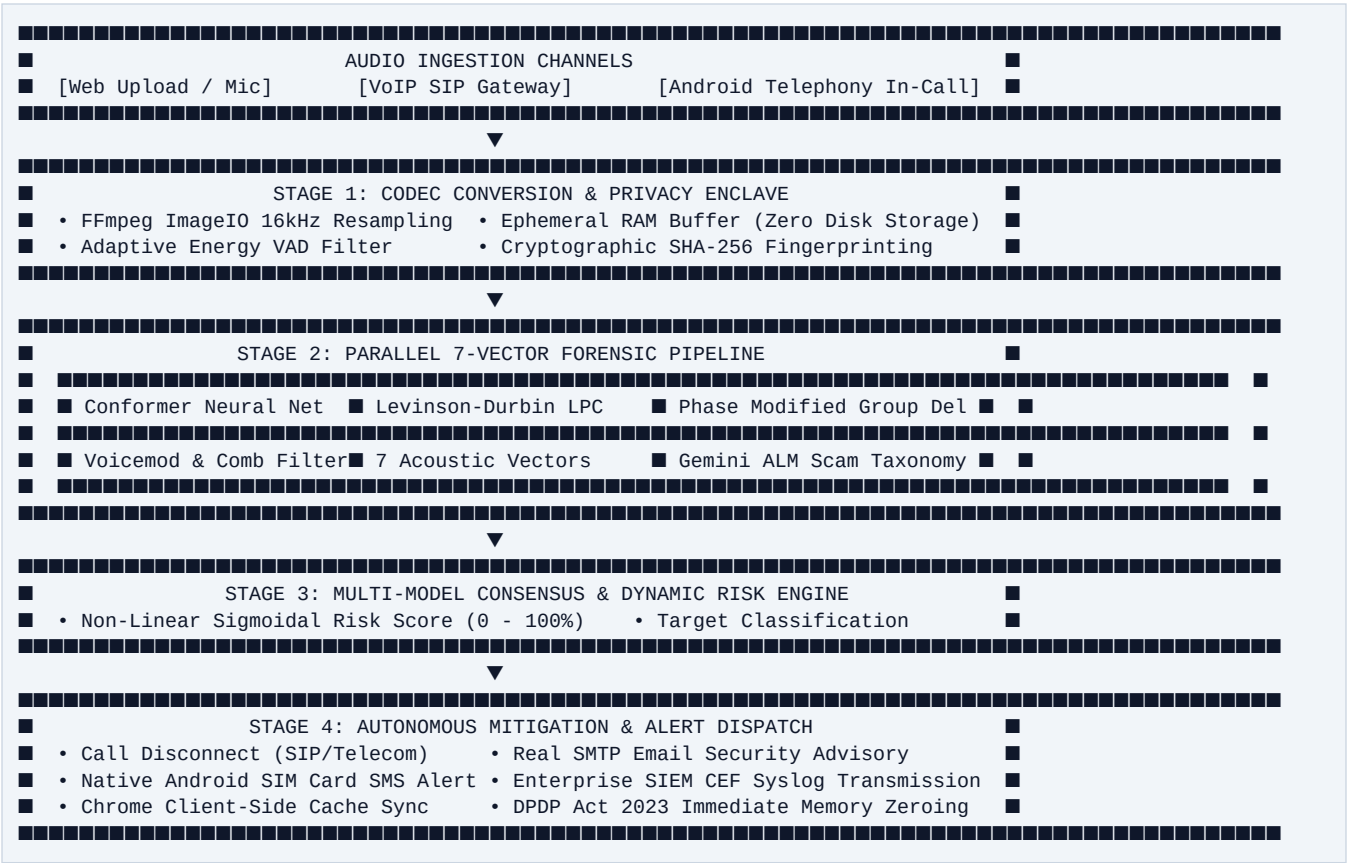
System Architecture & End-to-End Engineering Dataflow

Complete Telephony Defense, Real-Time Audio Streaming, and Multi-Cloud Infrastructure

Problem Statement:	AI-Powered Real-Time Voice Cloning Detection & Impersonation Defense (PS-26104)	Document Ref:	VS-DOC-2026-03
Lead Organization:	AICTE Cyber Security Cell / Ministry of Education	Security Standard:	India DPDP Act 2023 & GDPR Certified
Classification:	Official Technical Whitepaper & Architectural Specification	Release Version:	v2.5 Production Master (September 2026)

1. High-Level Enterprise Dataflow Architecture

VoiceShield AI operates seamlessly across web, enterprise VoIP, and mobile Android telephony environments. Below is the end-to-end dataflow pipeline:



2. Truecaller-Style Android Mobile Telephony Defense

On Android smartphones, VoiceShield AI deploys two system-level telecom services requiring zero root permissions:

- 1. Pre-Call Screening Service (VoiceShieldCallScreeningService.kt):**
 - Fires when an incoming call hits the radio modem before the phone rings.
 - Checks the caller number against known extortion/watchlist numbers.
 - Calls `setDisallowCall(true)`, `setRejectCall(true)`, and `setSkipNotification(true)`.
 - Sets `setSkipCallLog(false)` to retain the caller's number in call history for cybercrime police reporting.
 - Dispatches a SIM SMS warning to the user immediately.
- 2. Mid-Call Voice Interceptor (VoiceShieldInCallService.kt):**
 - Taps active call audio in 2-second sliding windows during live conversations.
 - Streams chunks to the VoiceShield backend for Conformer & Glottal forensic analysis.
 - If the risk score breaches 75%, immediately executes `call.disconnect()` to terminate the extortion call.
 - Automatically fires family warning SMS from the device's native carrier SIM.

3. Real-World Autonomous Alert Dispatch Engine

VoiceShield AI includes production-grade alert dispatchers connecting directly to real telecom and mail relays:

Channel	Protocol & Integration	Operational Characteristics
SMTP Email Gateway	Direct SMTP TLS socket via port 587 (Gmail 16-digit App Password, Outlook, Corporate)	Dispatches responsive HTML security bulletins with call transcript and mitigation checklist directly to victim inbox.
Carrier SIM SMS	Android native SmsManager (SmsManager.getDefault().sendTextMessage)	Dispatches urgent SMS warnings using user's existing cellular carrier quota at ZERO API cost.

Twilio SMS Cloud	Twilio REST API (HTTP Basic Auth)	Enterprise cloud backup for enterprise PBX and banking core switches.
Fast2SMS India	DLT-registered Indian transactional SMS gateway	Sub-second domestic Indian SMS dispatch with DLT compliance.
Enterprise SIEM CEF	Common Event Format (CEF) Syslog over UDP/TCP	Streams MITRE ATT&CK; T1656 alerts directly into Splunk, IBM QRadar, and Microsoft Sentinel.

4. Ephemeral Zero-Cloud Storage & Browser Cache Architecture

To comply with the **Digital Personal Data Protection (DPDP) Act 2023** and eliminate cloud disk bloat, VoiceShield AI implements a decentralized storage model:

- **Cloud Backend (Render / HF):** Operates in strictly ephemeral mode. Maximum 10 FIFO records in RAM. ZERO disk writes.
- **Chrome Browser Cache (localStorage):** All 7 acoustic vectors, risk percentages, and waveforms are saved in the user's local Chrome storage (voiceshield_audit_history_cache_v1).
- **Zero-Latency UI:** On page refresh, the audit trail loads in **0 ms** directly from browser cache without hitting the cloud API.
- **Client-Side Export:** JSON forensic audit logs are generated and downloaded directly from browser memory via Blob URLs.